

EpiParam

User's Guide

English Version

Contents

Introduction.....	3
Incubation period.....	5
Estimation using the conventional parametric models.....	7
Estimation using the Bayesian parametric models.....	10
Serial interval	12
Estimation using the conventional parametric models.....	14
Estimation using the Bayesian parametric models.....	16
Kappa (K) and R_0	18

Introduction

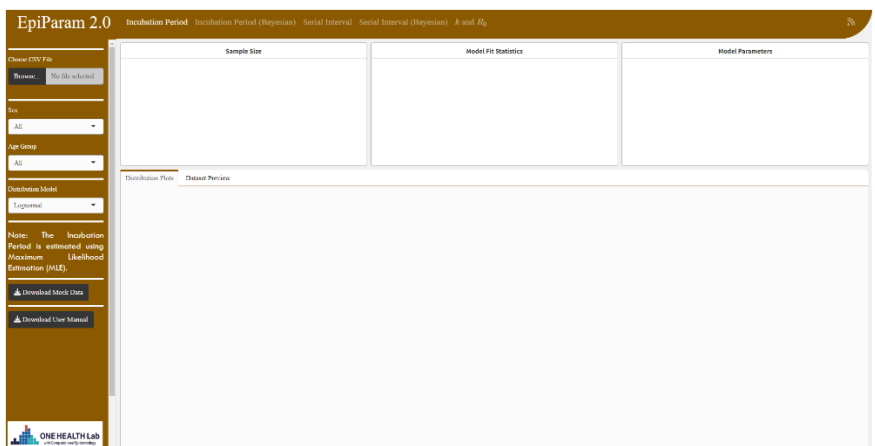
EpiParam (Epidemiological Parameter Estimation for Infectious Diseases) is a Shiny interface for estimating five main parameters (incubation period, serial interval, R_0 , R_t , and k) in the epidemiology of infectious diseases, developed by the OneHealth laboratory of the Department of Preventive Medicine at The Catholic University of Korea, Seoul, Korea. EpiParam introduces the Bayesian parametric model component that estimates the incubation period and serial interval when the exposure date and dates of onsets are intervals.

This user's guide provides instructions on how to use the EpiParam application to generate the incubation period, serial interval, using the conventional parametric models (lognormal, Weibull, gamma and normal) and the Bayesian parametric models. It also demonstrates the estimation of the reproduction number (R_0), and the Kappa value (k) for a given infectious disease.

EpiParam is published in the github repository <https://github.com/ACHANGWA/Epiparam>

To begin, open the GitHub repository and download all the supporting files into a folder on your desktop. Open the RMD file ([Epiparam Full App w Bayesian 02-02-2026.Rmd](#)) and run it in Rstudio in your computer.

Note that for a smooth functioning of the application, all the downloaded files from the repository must be place in a single folder in your computer.



This application was built to estimate five parameters: The incubation period, serial interval, reproduction number (R_0), and the Kappa value (k). It estimates the

incubation period and serial intervals using both the conventional parametric models and the Bayesian models.

Incubation period

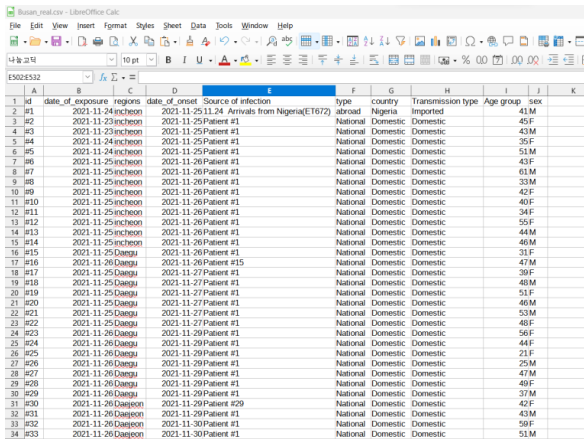
This is the period between exposure/infection of an individual by a pathogen and the manifestation of the signs and symptoms of the disease. To estimate the incubation period, click on the ‘Incubation period’ window in the application.

Data preparation.

1. Save your data spreadsheet in a ‘.csv’ format
example: “Busan_data.csv”
2. In the CSV data spreadsheet, the following variables should be present with the given specificities (Capital/small letter sensitive);
 - ✓ *Date of exposure of the disease → rename as “date_of_exposure”
 - ✓ *Date of symptom onset → rename as “date_of_onset”
 - ✓ Gender/Sex → rename as “sex”

- ✓ Age/Age-group → rename as “age_group”
- ✓ Any other variable depending on the researchers’ interests.

Mock data for the COVID-19 outbreak.



A	B	C	D	E	F	G	H	I	J	K
1	id	date_of_exposure	regions	date_of_onset	Source of infection	type	country	Transmission type	Age group	sex
2	i1	2021-11-24	incheon	2021-11-25	11.24 Arrivals from Nigeria(E167)	abroad	Nigeria	Imported	41M	
3	i2	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	45F	
4	i3	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	43M	
5	i4	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	35F	
6	i5	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	51M	
7	i6	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	43F	
8	i7	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	61M	
9	i8	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	33M	
10	i9	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	42F	
11	i10	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40F	
12	i11	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	34F	
13	i12	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	55F	
14	i13	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	44M	
15	i14	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	46M	
16	i15	2021-11-25	Daejeon	2021-11-26	Patient #1	National	Domestic	Domestic	31F	
17	i16	2021-11-26	Daejeon	2021-11-26	Patient #15	National	Domestic	Domestic	47M	
18	i17	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	39F	
19	i18	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	48M	
20	i19	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	51F	
21	i20	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	46M	
22	i21	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	53M	
23	i22	2021-11-25	Daejeon	2021-11-27	Patient #1	National	Domestic	Domestic	48F	
24	i23	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	56F	
25	i24	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	44F	
26	i25	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	21F	
27	i26	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	25M	
28	i27	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	47M	
29	i28	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	49F	
30	i29	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	37M	
31	i30	2021-11-26	Daejeon	2021-11-29	Patient #29	National	Domestic	Domestic	42F	
32	i31	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	43M	
33	i32	2021-11-26	Daejeon	2021-11-29	Patient #1	National	Domestic	Domestic	59F	
34	i33	2021-11-26	Daejeon	2021-11-30	Patient #1	National	Domestic	Domestic	51M	

This dataset can be downloaded from the ‘Download Mock data’ icon on the incubation period page after running the RMD.

Required formats

❖ date_of_exposure

This is the probable date of exposure of the case, in the format “YYYY-MM-DD”

❖ date_of_onset

This is the probable date of symptom onset of the disease, in the format “YYYY-MM-DD”

❖ sex

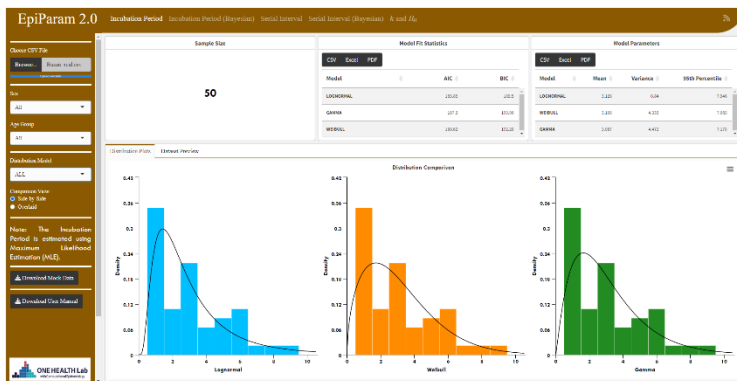
the gender of the individuals; All = Everyone, regardless of gender; Male = M; Female = F

❖ age

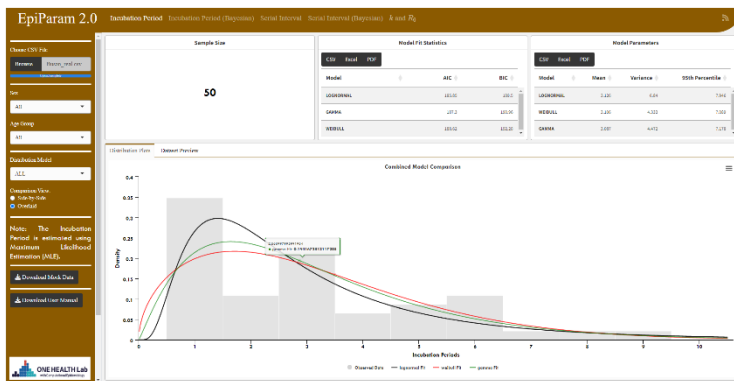
The ages of the participants should be numeric in years. Classifications of the age into age include; All is everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

A) Estimation using the conventional parametric models

To estimate the incubation period.



1. Click on the 'Incubation Period' icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on "Browse" and selecting the data from the computer.
3. Select which parametric model you wish to use (lognormal, Weibull, gamma or ALL)
4. In case you select 'ALL', select the comparison view. The view can either be 'side by side' or 'overlaid'.



Output Description

The output window is made up of 5 sub-windows.

- The ‘Sample Size’ window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The ‘Dataset Preview’ window, previews the uploaded dataset.
- The ‘Model Fit Statistics’ window that produces the model AIC and BIC.
- The ‘Model Parameters’ window, provides estimates (mean, variance, 95th percentile using the respective models.)

- The ‘Distribution plots’ window, shows the output histogram of the distributions.

Sub-group analysis

Based on individual needs, select stratifications by sex or by age group or a combination of both.

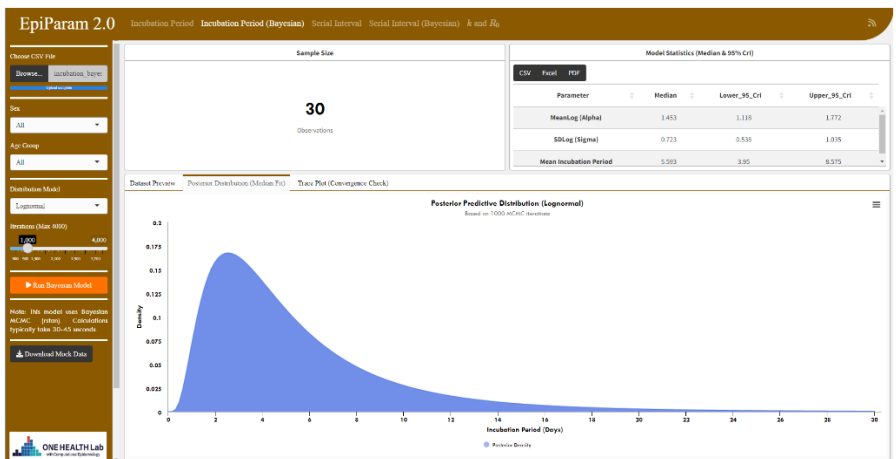
B) Estimation using the Bayesian parametric models

The Bayesian parametric models are used to estimate the incubation period when there exists an uncertainty in the exposure dates resulting in the dates being an interval.

- To estimate the incubation period using this method, click on the ‘Incubation Period (Bayesian)’ window and download the mock dataset at the bottom of the page.
- For this method, check that the dataset has the minimum and maximum Exposure dates for each entry
- Upload the dataset and choose the distribution (lognormal, Weibull, and gamma) of interest.

- Select the number of iterations (between 500 to 4000) and click on 'Run Bayesian Model' to run the model.

Note: The model uses Bayesian MCMC (rstan). And calculations typically take between 30-45 seconds.



Serial interval (SI)

The serial interval is the time between successive cases in a disease chain of transmission. The serial interval is estimated from the interval between clinical onsets, in which case it is the 'clinical onset serial interval' when these quantities are observable.

Data preparation.

Similar to the incubation period;

- Save your data spreadsheet in a '.csv' format
example: "Busan_data.csv"
- In the csv data spread sheet, the following variables should be present with the given specificities (Capital/small letter sensitive);
 - ✓ *Date of exposure of the disease → rename as "date_of_exposure"
 - ✓ *Date of symptom onset → rename as "date_of_onset"
 - ✓ *Source of infection → rename as "source_of_infection"

- ✓ Gender/Sex → rename as “sex”
- ✓ Age/Age-group → rename as “age_group”
- ✓ Any other variable depending on researchers’ interests

Mock data for the COVID-19 outbreak.

Busan_real.csv - LibreOffice Calc

File Edit View Insert Format Styles Sheet Data Tools Window Help

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A	B	C	D	E	F	G	H	I	J	K
id	date_of_exposure	regions	date_of_onset	Source of infection	type	country	Transmission type	Age group	sex	
#1	2021-11-24	incheon	2021-11-25	11-24 Arrivals from Nigeria(E1672)	abroad	Nigeria	Imported	41M		
#2	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	45F		
#3	2021-11-23	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	43M		
#4	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	35F		
#5	2021-11-24	incheon	2021-11-25	Patient #1	National	Domestic	Domestic	51M		
#6	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	43F		
#7	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	61M		
#8	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	33M		
#9	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	42F		
#10	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	40F		
#11	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	34F		
#12	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	55F		
#13	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	44M		
#14	2021-11-25	incheon	2021-11-26	Patient #1	National	Domestic	Domestic	48M		
#15	2021-11-25	Daegu	2021-11-26	Patient #1	National	Domestic	Domestic	31F		
#16	2021-11-26	Daegu	2021-11-26	Patient #15	National	Domestic	Domestic	47M		
#17	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	39F		
#18	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	48M		
#19	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	51F		
#20	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	46M		
#21	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	53M		
#22	2021-11-25	Daegu	2021-11-27	Patient #1	National	Domestic	Domestic	48F		
#23	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	56F		
#24	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	44F		
#25	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	21F		
#26	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	25M		
#27	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	47M		
#28	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	49F		
#29	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	37M		
#30	2021-11-26	Daegu	2021-11-29	Patient #29	National	Domestic	Domestic	42F		
#31	2021-11-26	Daegu	2021-11-29	Patient #1	National	Domestic	Domestic	43M		
#32	2021-11-26	Daegu	2021-11-30	Patient #1	National	Domestic	Domestic	59F		
#33	2021-11-26	Daegu	2021-11-30	Patient #1	National	Domestic	Domestic	51M		

Required formats

❖ date_of_exposure

This is the probable date of exposure of the case,
in the format “YYYY-MM-DD”

❖ date_of_onset

This is the probable date of symptom onset of the disease, in the format “YYYY-MM-DD”

❖ id as id. A numeric identity which is given to each data unit

❖ source_of_infection

The most probable source of infection of an individual given rise to the ‘infector – infectee’ pair

❖ sex

the gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females

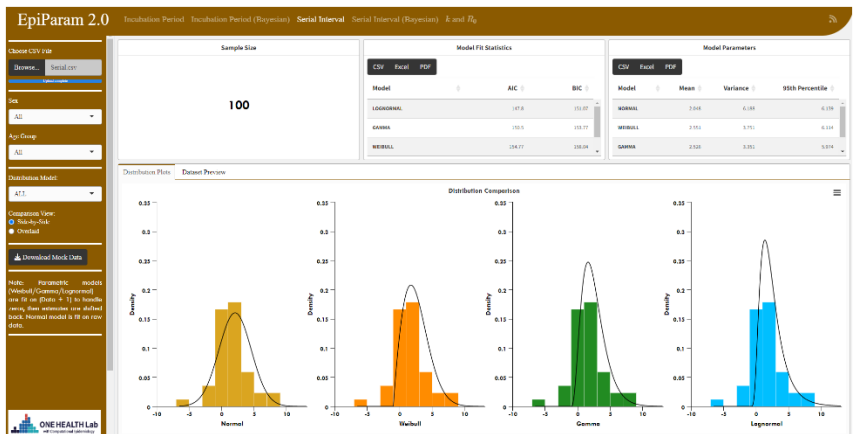
❖ age_group

The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

A) Estimating the Serial Interval using parametric models

In order to estimate the serial interval;

1. Click on the ‘serial interval’ icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from the computer.
3. This will produce an output as follows;



Output Description (SI)

The output window is made up of 5 sub-windows.

- The 'Sample Size' window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The 'Dataset Preview' window previews the uploaded dataset.
- The 'Model Fit Statistics' window that provides the AIC and BIC of each model.
- The 'Model Parameters' window that provides estimates (mean, variance, 25th, 75th, and 95th quantile) for each model.
- The 'Data Plots' window that demonstrates the output histogram of the distribution.

Sub-group analysis

Based on the researcher's needs, select stratifications by sex or by age group, or a combination of both.

B) Estimating the Serial Interval using the Bayesian parametric models

In order to estimate the serial interval using the Bayesian method;

1. Click on the ‘Serial Interval (Bayesian)’ icon at the top of the App.
2. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from the computer. The data here should contain the minimum and maximum onset dates of the infectors and infectees.
3. Upload the dataset and choose the distribution (lognormal, Weibull, normal and gamma) of interest.
4. Select the number of iterations (between 500 to 4000) and click on ‘Run Bayesian Model’ to run the model.

Note: The models estimates the serial interval by linking the infectors to the infectees. Therefore, ensure that ‘Source of infection’ IDs match the id. Calculations typically take between 30-45 seconds.

Kappa (K) and R_0

In the context of understanding and modelling transmission dynamics, the term "kappa" is sometimes used to represent the degree of variation in infectiousness among infected individuals. This variation in infectiousness is essential to understanding how different individuals may contribute differently to the spread of the disease.

Interpretation

A higher kappa value suggests greater variation, meaning that some individuals are highly infectious (super spreaders), while others have lower infectiousness. A lower kappa value implies less variation, where most individuals have similar levels of infectiousness.

Note: The "kappa" and its associated values can vary depending on the modelling approach. The range of kappa values typically varies from 0 to 1, where:

Kappa = 0 - Implies no variation in infectiousness among individuals. All infected individuals are equally infectious.

Kappa = 1 - Indicates the highest level of variation in infectiousness, where there is extreme heterogeneity among infected individuals.

Data preparation for parameter estimation

A) Save your data spreadsheet in a '.csv' format

B) In the CSV data spreadsheet, the following variables should be present in your dataset;

- ✓ Age - Age of the participants as whole numbers
- ✓ Gender- Gender of the participants
- ✓ Date of onset - Date of onset of symptoms for the respective disease
- ✓ Date of confirm – Date of laboratory confirmation for the respective disease
- ✓ Source of infection ID – The ID number of the individual from which the infection was gotten.
- ✓ Cluster_id – Assigned ID of the infection cluster (A,B,C... etc or any other assigned style)

- ✓ Cluster character – Name or label that is given to the cluster
- ✓ Infectee generation – the number of successive generation of infections from a primary case in a specific cluster.

Note: A mock dataset has been provided for your information. You can download the mock data in the ‘Download Mock data’ tab in the app.

Required formats

- ❖ Date of confirm – Date of laboratory confirmation in the format “DD-MM-YYYY”
- ❖ Date of onset - This is the probable date of symptom onset of the disease, in the format “DD-MM-YYYY”
- ❖ Gender
The gender of the individuals; All = Everyone, regardless of gender; Male = Male; Female = Females
- ❖ Age

The ages of the participants in years and should be imputed as a whole number. Classifications of the age into age groups include; All = Everyone regardless of age; 0-19, 20-29, 30-39, 40-49, 50-59, and ≥ 60

Estimating the Kappa (K) and R_0

To estimate the K and R_0 ;

5. Click on the ‘K and R_0 ’ icon at the top of the App.
6. Upload the .csv file containing the cleaned data you wish to use, by clicking on “Browse” and selecting the data from your device.
7. This will produce an output as follows;

The screenshot displays the EpiParam 2.0 interface. On the left, there's a sidebar with 'Cluster Size' set to 64. The main area is divided into three sections: 'Sample Size' (showing 64), 'Model Parameters' (with a table for parameters like R_0 and K), and 'Confidence Interval' (showing intervals for R_0 and K). Below these is a table of infection data with columns for ID, age, gender, dates, source, transmission relationship, cluster ID, character, generation, and infection status. The table shows 10 rows of data, including details like 'unlabeled', 'social', and 'family' transmission types.

id	age	gender	date_of_onset	date_of_confirm	source_of_infection_id	transmission_relationship	cluster_id	cluster_character	infection_generation	infectogen1	infectogen2	infectogen3
9	0	F	2020-05-01	2020-05-05	#1	#1 → #0	0	social	1	1	0	
10	12	F	2020-05-03	2020-05-06	unlabeled	400	0	social	0	10		
14	6	M	2020-05-05	2020-05-06	#1	#1 → #0	0	social	1	1	1	14
28	9	F	2020-05-05	2020-05-06	#1	#1 → #0	0	social	1	1	1	28
37	12	F	2020-05-08	2020-05-09	unlabeled	837	0	social	0	37		
42	11	M	2020-05-08	2020-05-09	#11	#11 → #42	0	social	1	11	42	
43	10	M	2020-05-10	2020-05-10	#12	#12 → #43	0	family	2	43	35	43
61	0	M	2020-05-12	2020-05-13	#02	#02 → #61	0	social	1	62	61	
68	0	M	2020-05-13	2020-05-13	#21	#21 → #68	0	social	1	73	68	

Output Description

The output window is made up of 7 sub-windows.

- The 'Sample Size' window, gives the total number of rows (individuals/patients/units) in the uploaded dataset.
- The 'Dataset preview' window previews the uploaded dataset.
- The 'Model parameter' window that produces estimated K and R_0 values as well as the corresponding standard deviations (SD).
- The 'Confidence Interval' window provides the confidence interval of each of the parameters.
- 'Histogram window' demonstrates the output histogram of the distribution.
- The 'Transmission relationship' window demonstrates the transmission relationship of each cluster. This can be viewed by selecting the cluster ID in the dropdown section of the transmission relationship window, as shown below;

